

IIoT Lab

Interfacing Arduino with Node-RED to Monitor Temperature and Humidity on a Webpage

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Hardware: Arduino Uno R3, DHT11 Sensor, Jumper Wires, USB A-B Cable

Software: Arduino IDE, Node.js, Node-RED

Objective

The aim of this experiment is to interface an Arduino Uno R3 with Node-RED using a DHT11 temperature and humidity sensor. Sensor data is transmitted via serial communication, processed by a Node-RED function node, and displayed as live gauge widgets on a web dashboard.

Required Node Packages

The following two packages must be installed before building the Node-RED flow:

Package	Purpose	Install Command
node-red-dashboard	Gauge and UI widgets for the browser dashboard	npm install node-red-dashboard
node-red-node-serialport	Serial In/Out nodes to read from Arduino's COM port	npm install node-rednode-serialport

Alternatively, go to Node-RED → ■ Menu → Manage Palette → Install tab, then search and install each package.

Step 1 — Arduino Setup

1.1 Hardware Connection

Connect the DHT11 sensor to the Arduino Uno R3 as follows:

DHT11 Pin	Arduino Pin
VCC	5V or 3.3V
GND	GND
DATA	Digital Pin 2

1.2 Arduino Code

Install the **DHT sensor library by Adafruit** from the Arduino IDE Library Manager (Sketch → Include Library → Manage Libraries). Then upload the following code:

```

#include <DHT.h>

#define DHTPIN  2

#define DHTTYPE  DHT11

DHT      dht(DHTPIN,
DHTTYPE);

void setup() {
  dht.begin();
  Serial.begin(9600); }

void loop() {
  float H = dht.readHumidity();  float T
= dht.readTemperature();

  if (isnan(H) || isnan(T)) {    Serial.println("Failed to read
from DHT sensor!");    return;  }

  Serial.print("Humidity:"); Serial.print(H);
  Serial.print(",Temp:");  Serial.println(T);

  delay(2000);
}

```

The code reads humidity and temperature every 2 seconds and sends them over serial in the format:
Humidity:66.00,Temp:33.80

Step 2 — Node-RED Flow Setup

Open Node-RED at <http://127.0.0.1:1880> and create a new flow using the following nodes:

2.1 Nodes Required in the Flow

Node	Configuration
Serial In	Port: COM6 (or your port), Baud rate: 9600. Name it 'Arduino Uno R3'.

Function	Parses the serial string and outputs temperature and humidity separately.
Gauge (×2)	One for Temperature (0–50°C), one for Humidity (0–100%).
Debug	Connected to function output to verify values in the debug panel.

2.2 Function Node Code

Double-click the Function node and paste the following code in the 'On Message' tab:

```
var raw = msg.payload;

if (Buffer.isBuffer(raw)) { raw = raw.toString(); } raw =
raw.trim();

var humMatch = raw.match(/Humidity:\s*([\d.]+)/); var
tempMatch = raw.match(/Temp:\s*([\d.]+)/);

if (!humMatch || !tempMatch) {
  node.warn("Could not parse: " + raw);
  return null; }

var hum = parseFloat(humMatch[1]); var
temp = parseFloat(tempMatch[1]);

// Output 1 → Temperature gauge | Output 2 → Humidity gauge return [{
payload: temp }, { payload: hum }];
```

2.3 Gauge Node Settings

Setting	Temperature Gauge	Humidity Gauge
Title	Temperature	Humidity
Value Format	{{value}} °C	{{value}} %
Min / Max	0 / 50	0 / 100

2.4 Flow Diagram

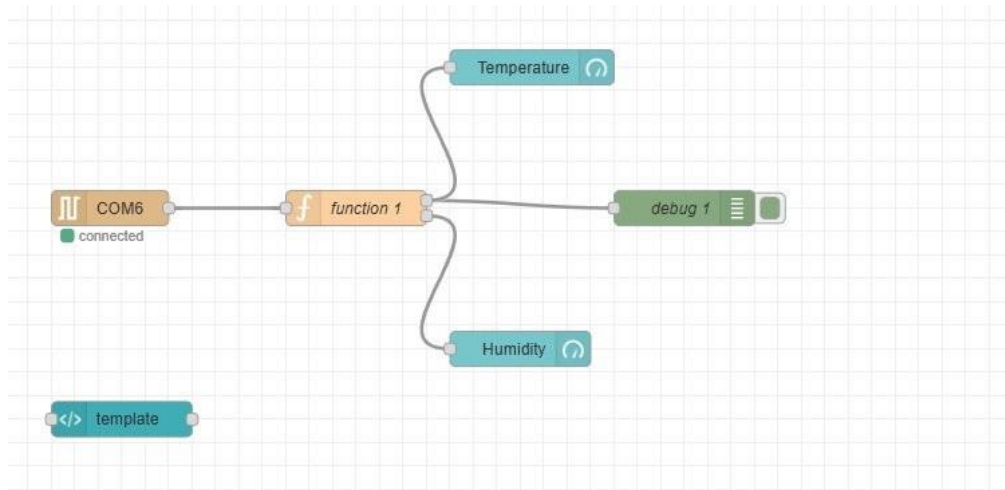


Figure 1: Node-RED flow — Serial In (COM6) → Function → Temperature Gauge / Humidity Gauge + Debug

Step 3 — Deploy and Test

1. Wire the nodes: Serial In → Function → Temperature Gauge (output 1), Humidity Gauge (output 2), Debug.
2. Click the red **Deploy** button in the top-right corner of Node-RED.
3. Open a browser and go to **<http://127.0.0.1:1880/ui>**.
4. You should see live temperature and humidity readings displayed as gauge widgets.

Dashboard Output:

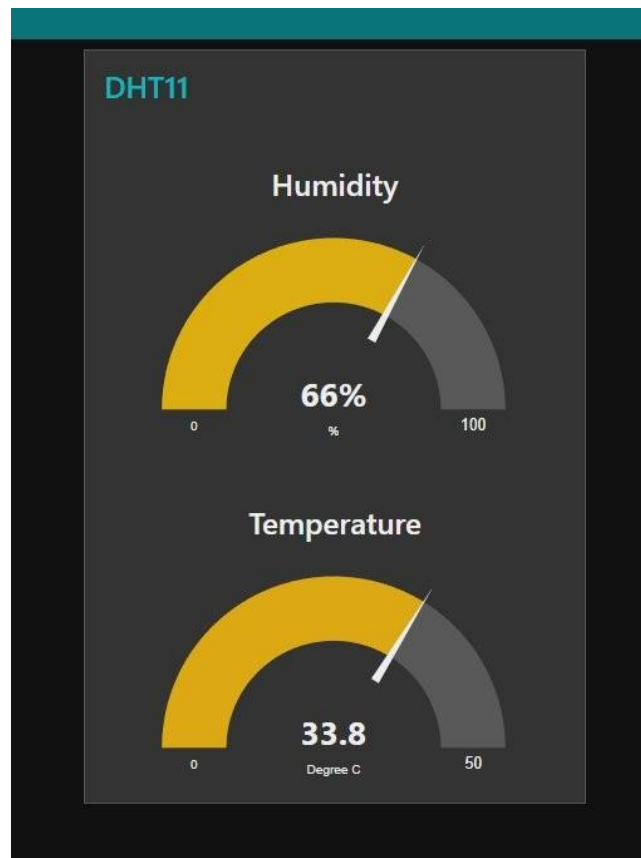


Figure 2: Live dashboard showing Humidity: 66% and Temperature: 33.8°C

Optional — Custom CSS via Template Node

To customise the dashboard appearance, add a **Template** node to the flow and paste the CSS below. This changes the background and font styling of the dashboard.

```
<style> body, md-content {  
    background-color: #1a1a2e !important; font-  
family: 'Segoe UI', sans-serif; } .nr-dashboard-  
theme .md-toolbar-tools { background-color:  
#16213e !important; } .nr-dashboard-theme  
.gauge-label { color: #e0e0e0 !important;  
font-size: 16px; font-weight: bold; } .nr-  
dashboard-theme .gauge-value { fill: #00b4d8  
!important; font-size: 26px !important; }  
.nr-dashboard-theme .nr-dashboard-widget {
```

```
background: #0f3460 !important; border-  
radius: 8px; padding: 10px; }  
</style>
```

To apply: drag a Template node → paste the CSS → set type to 'Widget in group' → Deploy → refresh the dashboard URL.

Conclusion

This experiment successfully demonstrated the integration of an Arduino Uno R3 with Node-RED for real-time IoT monitoring. The DHT11 sensor transmitted temperature and humidity values via serial communication at 9600 baud. A function node in Node-RED parsed the incoming data and routed each value to a dedicated gauge widget on the web dashboard. The result was a complete IIoT data pipeline — from physical sensor to web-based display — built entirely using open-source tools.